

Numerical Implementation and Reproducibility

A Nonlocal Attraction–Repulsion Transport Equation

This document describes the supplied `matlabcode.m` script, in the order of the numerical examples in the manuscript: one dimension followed by two dimensions. The script itself executes the two-dimensional example first. Its numerical algorithms and parameters have not been changed in this package. Descriptions below refer to the actual implementation, including normalization, truncation, and fallback procedures.

1 Running the code

Place `matlabcode.m` in a writable MATLAB working directory and run

`matlabcode`

The script clears workspace variables, closes open figures, and clears the command window. It contains its own local functions and needs no external input data. It uses `hypergeom` for the Green-kernel evaluation; a MATLAB installation providing this function is required (normally through Symbolic Math Toolbox). It uses `lsqnonneg` when available and supplies a projected-gradient fallback otherwise. It also calls `fzero`, `fminbnd`, `besselj`, `quantile`, and plotting functions including `xline`, `yline`, and `sgtitle`. A recent MATLAB installation supporting local functions in scripts and these plotting functions is appropriate. The source does not record the MATLAB release or toolbox versions used for the original figures.

The generated files are

- `fig_1d_a_greater_r_phi_compare.pdf` and its PNG counterpart;
- `fig_2d_ar_phi_compare.pdf` and its PNG counterpart.

Export uses `exportgraphics`, with vector content for PDF and 220 dpi for PNG. If export fails, the script uses `print`. Existing outputs with the same names may be overwritten. Console output includes masses, residuals, CDF discrepancies, and the one-dimensional final diameter. This document records the algorithm by source inspection; it does not report newly executed MATLAB results.

2 One-dimensional experiment

2.1 Parameters and stationary Riesz equation

The background and exponents are

$$\omega_m = \frac{m}{2} \mathbf{1}_{[-1,1]}, \quad (a, r, m) = (0.8, 0.2, 1.8).$$

For a trial support $[-L, L]$, partition the interval into n uniform cells $[e_{j-1}, e_j]$ of width $h = 2L/n$, with midpoints x_j . Write $\phi_h = \sum_j c_j \mathbf{1}_{[e_{j-1}, e_j]}$. Midpoint collocation gives

$$Ac = b, \quad A_{ij} = r(1+r) \int_{e_{j-1}}^{e_j} |x_i - y|^{r-1} dy,$$

$$b_i = \frac{m(1+a)}{2} [\operatorname{sgn}(x_i + 1)|x_i + 1|^a - \operatorname{sgn}(x_i - 1)|x_i - 1|^a].$$

Each cell integral, including the diagonal singular cell, is evaluated analytically:

$$\int_{\ell}^u |x - y|^{r-1} dy = \begin{cases} ((x - \ell)^r - (x - u)^r)/r, & x \geq u, \\ ((u - x)^r - (\ell - x)^r)/r, & x \leq \ell, \\ ((x - \ell)^r + (u - x)^r)/r, & \ell < x < u. \end{cases}$$

MATLAB's backslash operator solves the linear system. The support is selected by `fzero` applied to $M_{140}(L) - 1$ on $[0.16, 0.45]$, where $M_n(L) = h \sum_j c_j$. No custom solver tolerances are specified. At the selected L , the system is recomputed with $n = 320$; the support root is not recomputed on this finer grid. The script prints its mass and relative algebraic residual

$$\mathcal{R}_{\text{Riesz}} = \|Ac - b\|_2 / \|b\|_2.$$

The density curve uses the raw coefficients c_j , without clipping. The cumulative distribution is constructed at cell edges by integration and divided by its final mass. Thus CDF normalization is distinct from the reported mass of the computed density.

2.2 Exterior Poisson and fractional-Laplacian reconstruction

This calculation uses the prescribed half-width $L_P = 0.2727$, $A = -L_P$, $B = L_P$, $n = 320$ interior cells, and $\sigma = r/2 = 0.1$. At interior midpoints set $g_i = b_i/[r(1+r)]$. For $y \notin [A, B]$,

$$\tilde{g}_h(y) = h \sum_i P(y, x_i) g_i, \quad P(y, x) = \frac{\sin(\pi\sigma)}{\pi} \left[\frac{(y - A)(y - B)}{(x - A)(B - x)} \right]^{\sigma} \frac{1}{|y - x|}.$$

The denominator $(x - A)(B - x)$ is floored at MATLAB's `eps`. The exterior-domain kernel above is the one implemented in this version.

On each side, the exterior integral uses $n_o = 1400$ nodes, $Y_{\max} = 10$, and grading exponent $\beta = 1.8$:

$$u_j = \frac{j - 1/2}{n_o}, \quad z_j = (Y_{\max} - L_P) u_j^{\beta} + 10^{-6}, \quad w_j = \frac{(Y_{\max} - L_P)^{\beta}}{n_o} u_j^{\beta-1},$$

with $y_j^- = A - z_j$ and $y_j^+ = B + z_j$. Up to a common multiplicative constant, the reconstructed values are

$$\begin{aligned} f_i &= h \sum_{\ell \neq i} \frac{g_i - g_{\ell}}{|x_i - x_{\ell}|^{1+2\sigma}} \\ &\quad + \sum_j \frac{g_i - \tilde{g}_h(y_j^-)}{|x_i - y_j^-|^{1+2\sigma}} w_j + \sum_j \frac{g_i - \tilde{g}_h(y_j^+)}{|x_i - y_j^+|^{1+2\sigma}} w_j. \end{aligned}$$

Interior terms at distance at most 10^{-12} are excluded. No correction for the omitted exterior tail is added. The code computes $M_- = h \sum_i \max(-f_i, 0)$ and prints a diagnostic if $M_- > 10^{-4}$. It then clips and normalizes:

$$\hat{f}_i = \frac{\max(f_i, 0)}{h \sum_{\ell} \max(f_{\ell}, 0)}.$$

If the denominator is nonpositive, a warning is issued and the normalized nonnegative part of g is used instead. The omitted common multiplicative constant cancels in this normalization. The resulting CDF is normalized once more by its final cumulative mass.

The finite-Riesz CDF is linearly interpolated to the Poisson reconstruction's cell edges, with linear extrapolation if necessary. The reported discrepancy is the maximum absolute difference on those edges. It compares normalized CDFs on two potentially different support grids.

2.3 Particle dynamics and diagnostics

The particle system has $N = 500$, time step $\Delta t = 0.02$, and 2500 explicit Euler steps, giving $T = 50$. With `rng(12)`, its initial positions are

$$X_i^0 = -1.4 + \frac{2.8(i-1)}{N-1} + 0.01\xi_i,$$

where ξ_i are the values returned by `randn`. Initial positions are sorted. The exact background force is

$$F_\omega(x) = \frac{m}{2} (|x+1|^{1+a} - |x-1|^{1+a}),$$

and the update is

$$X_i^{k+1} = X_i^k + \Delta t \left[-F_\omega(X_i^k) + \frac{1+r}{N} \sum_j \text{sgn}(X_i^k - X_j^k) |X_i^k - X_j^k|^r \right].$$

All pairwise terms use the old positions. The diagonal is explicitly set to zero. No force smoothing is used. Positions are sorted every 20 steps and after the final step. Termination is at the fixed time T , with no residual-based early stopping.

Every four steps the script records the diameter $D = \max_i X_i - \min_i X_i$ and twice the MATLAB 99% quantile of $|X_i|$. These are plotted with $2L$. For the empirical CDF $F_N(x, T) = N^{-1} \sum_i \mathbf{1}_{\{X_i(T) \leq x\}}$ and normalized Riesz CDF F_h , the reported error is

$$\mathcal{E}_{\text{CDF}} = \max_{0 \leq j \leq 320} |F_N(e_j, T) - F_h(e_j)|.$$

This is a maximum on the selected cell edges, not an independently computed supremum over all real x . The histogram uses 55 equally spaced edges (54 bins) on $[-1.05L, 1.05L]$ and `Normalization='pdf'`.

3 Two-dimensional radial experiment

3.1 Radial kernel and stationary discretization

Here

$$(a, r, m) = (0.6, 0.2, 1.8), \quad \omega_m(x) = \frac{m}{2\pi} e^{-|x|^2/2}, \quad R = 0.33.$$

The radius R is fixed in the script. A source comment attributes this choice to a coarse residual scan, but that scan is not implemented in the supplied file. The radial force of a unit circular mass is approximated as

$$\mathcal{A}_{\alpha,q}(\rho, s) = \frac{1+\alpha}{q} \sum_{\ell=1}^q (\rho - s \cos \theta_\ell) [\max(\rho^2 + s^2 - 2\rho s \cos \theta_\ell, 10^{-14})]^{(\alpha-1)/2},$$

where $\theta_\ell = 2\pi(\ell - 1/2)/q$. The distance floor is applied before evaluating the power.

The stationary discretization uses 110 source rings at $s_j = (j - 1/2)R/110$, 150 test radii $\rho_i = (i - 1/2)R/150$, and $q = 160$ angular nodes. The background uses 550 radial midpoint nodes on $[0, 6]$:

$$\Delta s_b = 6/550, \quad s_\ell^b = (\ell - 1/2)\Delta s_b, \quad w_\ell^b = m s_\ell^b e^{-(s_\ell^b)^2/2} \Delta s_b.$$

These truncated weights are not renormalized. Set

$$A_{ij} = \mathcal{A}_{r,160}(\rho_i, s_j), \quad b_i = \sum_{\ell} \mathcal{A}_{a,160}(\rho_i, s_\ell^b) w_\ell^b.$$

The code solves

$$\min_{\mu \geq 0} \{ \|A\mu - b\|_2^2 + 120^2(\mathbf{1}^\top \mu - 1)^2 \}$$

with `lsqnonneg`, using default settings. This is midpoint force collocation with a nonnegative least-squares solve (called the radial Galerkin curve in the script). The result is clipped and divided by its total mass.

If `lsqnonneg` is unavailable, define $\tilde{A} = [A; 120\mathbf{1}^\top]$ and $\tilde{b} = [b; 120]$. Starting with the nonnegative part of the backslash solution, perform 800 iterations

$$\mu^{k+1} = \max\{\mu^k - \tau \tilde{A}^\top (\tilde{A}\mu^k - \tilde{b}), 0\}, \quad \tau = 0.9/(\|\tilde{A}\|_2^2 + 10^{-12}).$$

The final normalized solution defines $\text{Res} = A\mu - b$. The plot shows $|\text{Res}_i|$, and the console reports $\|\text{Res}\|_2/\sqrt{150}$ and $\|\text{Res}\|_\infty$. These are sampled stationary force residuals, not dynamical stopping criteria.

3.2 Green reconstruction and scalar calibration

Let $s = (1 + r)/2 = 0.6$. The source shape is evaluated using 2500 uniformly spaced frequencies k_ℓ from 10^{-5} to 16:

$$q_h(\rho) = \max \left\{ m\Delta k \sum_{\ell} k_\ell^{1+r-a} e^{-k_\ell^2/2} J_0(k_\ell \rho), 0 \right\}.$$

The computation suppresses overall Fourier normalization constants. Exterior quadrature uses 85 nodes with grading exponent 1.35:

$$u_j = (j - 1/2)/85, \quad \eta_j = R + (4 - R)u_j^{1.35}, \quad w_j = (4 - R)1.35u_j^{0.35}/85.$$

Angular averages use 96 midpoint nodes. For exterior ρ, η , define

$$d_\ell^2 = \max\{\rho^2 + \eta^2 - 2\rho\eta \cos \theta_\ell, 10^{-12}\}, \quad z_\ell = \frac{(\rho^2 - R^2)(\eta^2 - R^2)}{R^2 d_\ell^2}.$$

The kernel shape is

$$G_h(\rho, \eta) = \frac{1}{96} \sum_{\ell} (d_\ell^2)^{-(1-r)/2} I_s(z_\ell), \quad I_s(z) = \frac{z^s}{s} {}_2F_1(s, 1; s + 1; -z).$$

The auxiliary function $I_s(z) = \int_0^z t^{s-1}/(1+t) dt$ is evaluated by `hypergeom`, with negative arguments z clipped to zero before evaluation. Set

$$H_i = \sum_j G_h(\eta_i, \eta_j) q_h(\eta_j) 2\pi \eta_j w_j.$$

For interior radii s_i , the correction is

$$C_i = \sum_j H_j 2\pi \eta_j w_j \frac{1}{96} \sum_{\ell} [\max\{s_i^2 + \eta_j^2 - 2s_i \eta_j \cos \theta_\ell, 10^{-12}\}]^{-(3+r)/2}.$$

No tail correction beyond exterior radius 4 is included. With $\Delta s = R/110$, form

$$\nu_i(c) = \frac{\max\{q_h(s_i) + cC_i, 0\} 2\pi s_i \Delta s}{\sum_j \max\{q_h(s_j) + cC_j, 0\} 2\pi s_j \Delta s}.$$

The coefficient is selected by `fminbnd`, minimizing $\|A\nu(e^\ell) - b\|_2/\sqrt{150}$ for $\ell \in [-10, 10]$, with default settings. If the optimization throws an error, $c = 1$ is used. In the objective function only, nonpositive total mass triggers a uniform-mass fallback.

This coefficient changes the relative contributions of q_h and C ; it is not simply an overall normalization. Thus the displayed reconstruction includes a scalar fit to the same force equation used for the stationary ring masses. A fully independent evaluation with all constants fixed analytically would require a different implementation.

3.3 Radial evolution and visualization

The dynamics uses $N = 180$ equal-weight radial nodes, initialized deterministically by

$$\rho_i^0 = 1.1\sqrt{-2\log(1 - 0.98u_i)}, \quad u_i = (i - 1/2)/N.$$

The force tables use 260 uniform radii on $[0, 3.2]$ and 96 angular nodes. The background table uses the same 550 background nodes. Background queries use linear interpolation and linear extrapolation; self-force queries use bilinear interpolation with zero outside the table domain. Denote these interpolants by $\tilde{F}_\omega$ and $\tilde{\mathcal{A}}_r$. The explicit Euler update is

$$\rho_i^{k+1} = \max \left\{ \rho_i^k + \Delta t \left[-\tilde{F}_\omega(\rho_i^k) + \frac{1}{N} \sum_j \tilde{\mathcal{A}}_r(\rho_i^k, \rho_j^k) \right], 0 \right\}.$$

The diagonal ring contribution is retained. A radial node represents mass uniformly distributed over a circle, so this contribution differs from a point-particle diagonal force. Radii are sorted after every step. The time step is 0.05, and 1000 steps give $T = 50$. There is no residual-based early stopping.

Stationary cumulative masses $\sum_{s_j \leq \rho} \mu_j$ and $\sum_{s_j \leq \rho} \nu_j$ are evaluated on 500 radii in $[0, 0.66]$ and plotted against the empirical radial CDF $N^{-1} \sum_i \mathbf{1}_{\{\rho_i(T) \leq \rho\}}$. The script does not print a two-dimensional CDF error. The cloud visualization assigns angles

$$\vartheta_i = (i - 1)\pi(3 - \sqrt{5}) \pmod{2\pi}, \quad Z_i = \rho_i(T)(\cos \vartheta_i, \sin \vartheta_i).$$

These angles are for display only; Cartesian particle trajectories are not integrated.

4 Scope of the supplied reproducibility package

The code and this note specify the finite computations used to compare stationary profiles and evolved distributions. All numerical residual values must be obtained by running the script; none are invented here. The supplied script contains no systematic refinement study in particle number, time step, quadrature order, interpolation resolution, or exterior cutoff. Its 140-cell support solve followed by a 320-cell density solve is a two-stage calculation, not a reported refinement study. Comparisons of the two stationary constructions also do not replace a refinement study.

For a versioned release, retain this note and the exact script together. The package contains a SHA-256 checksum of the script for identifying the documented source. A MATLAB run log can additionally record `version`, `ver`, the console diagnostics, and whether the least-squares fallback was used. No execution log, convergence table, or repository DOI is claimed by this package.